

Payroll v4 (MGGO Payroll SOP v4.0) — request, ship, verify

Admin prod: <https://dpdadmin-prod.vercel.app/en/payroll> · DB: `eoksxkdssptgyqyywdju` · Migrations: `20261031200000`, `20261031200100`, `20261031200200`, `20261031200300`, `20261101200000` · Tests: `npm run test:payroll` = **120 / 120 pass** · Rider app: **no change required**

0. Surfaces, access and readiness

Surface	Route	Permission	Ready?
Payroll month, KPIs, export	<code>/en/payroll</code>	<code>payroll.view</code> · <code>export</code> <code>payroll.export</code>	Yes — Vercel prod
Combined grid + manual adjustments	<code>/en/payroll/combined</code>	<code>payroll.manage</code> to write	Yes
Attendance & Orders (read-only)	<code>/en/payroll/attendance-orders</code>	<code>payroll.view</code>	Yes
Requests queue	<code>/en/payroll</code> → Requests	<code>payroll.view</code>	Yes
Client rules, bands, simulator, change log	<code>/en/payroll/settings</code>	<code>payroll.manage</code> to save	Yes
Off Structure (per rider + bulk)	Header button on <code>/en/payroll</code>	<code>payroll.manage</code>	Yes
Rider app	—	—	No change · nothing to publish

Verified on production before this pack was written: tables `payroll_clients`, `payroll_client_rules`, `payroll_zone_metrics`, `payroll_zone_settings`, `payroll_manual_adjustments` exist; functions `admin_payroll_rule_snapshot`, `admin_apply_payroll_adjustments`, `admin_recompute_payroll_zone_metrics`, `admin_save_payroll_client`, `admin_save_payroll_client_rules`, `admin_save_payroll_zone_override`, `admin_save_payroll_zone_settings`, `admin_delete_payroll_client`, `admin_set_driver_off_structure`, `admin_payroll_month_snapshot` all present. No migration is pending.

1. What the client requested

- Move payroll from one hard-coded classification to a **client-owned rules engine** (MGGO Payroll SOP v4.0).
- Per client: **hour values** (full / half / reduced), required hours per day, a **default day status**, and Good / Average thresholds.
- Per client and per month: an **ordered rule list** matching on **zone band**, **zone name**, **orders** and **worked hours**; first match wins.
- The SOP result codes: 12 full day · 3h reduced · HALF half · ACT actual · ABS absent · ALH absent LH · ALO absent LO · CUS custom.
- Fixed precedence: **manual adjustment** > **Operations** > **first matching rule** > **client default** > **legacy fallback**.
- Zone efficiency from the **previous completed month only**: $DPD = \text{orders} \div \text{rider-days}$, target $DPD = \text{the average across active zones}$, $\text{efficiency} = DPD \div \text{target} \times 100$, banded Good ≥ 110 / Average ≥ 70 / Low < 70 .
- An **unmeasured zone is not "Low"**, and a zone with no rider-days must not drag the target down.
- Off Structure (contracted OFF days, default 2) driving Required Hours = $(\text{month days} - \text{OFF}) \times \text{full-day hours}$, with a per-rider editor and a **bulk Excel upload**.
- An Excel-like grid: block selection, fill, copy / paste, and one reason for a whole batch, recorded in an audit trail.
- A rule simulator so Ops can see the outcome of a hypothetical day without saving.
- All of it editable by staff without a code release.

2. What we shipped — request to implementation

Request	Shipped as	Where to see it
Client catalog + hour values	<code>payroll_clients</code> (uses <code>zone</code> / uses <code>orders</code> / uses <code>hours</code> , full / half / reduced hours, required hours per day, default off days, default result, Good & Average thresholds). <code>drivers.project_key</code> and <code>driver_intakes.project_key</code> now FK to it, so a client added in Settings is valid everywhere.	Settings → Client criteria
Ordered rule list per client per month	<code>payroll_client_rules</code> (<code>conditions</code> jsonb + <code>result</code> jsonb, unique on client + month + <code>sort_order</code>). A new month copies the previous month; the first month seeds the SOP rules.	Settings → Rules · <i><month></i>
One evaluation everywhere	Pure <code>payroll-rules-engine.ts</code> (<code>evalDay</code> , <code>firstMatchingRule</code> , <code>matchesCondition</code>) used by the grid, the CSV, the Attendance & Orders tab, the simulator and the unit tests. <code>admin_payroll_rule_snapshot</code> is one call for the whole grid.	Any tab · must agree cell for cell
Zone efficiency	<code>payroll_zone_metrics</code> + <code>admin_recompute_payroll_zone_metrics</code> ; Ops override of DPD / target / category through <code>admin_save_payroll_zone_override</code> , kept beside the automatic value.	Settings → Zones & efficiency
Zone settings per client	<code>payroll_zone_settings</code> + <code>admin_save_payroll_zone_settings</code> (global target, Good / Average, per-zone efficiency override).	Settings → Zones & efficiency
Manual adjustments + audit	<code>payroll_manual_adjustments</code> append-only, written by <code>admin_apply_payroll_adjustments</code> . Nothing is updated in place, so a correction is a new row.	Combined tab → Manual changes record
Off Structure bulk	<code>admin_set_driver_off_structure</code> + <code>off-structure-bulk.ts</code> preview (verdicts below). Only real changes are written; default OFF = 2 stays implicit.	Header → Off Structure
Rule change audit	<code>payroll_rule_audit_logs</code> + <code>payroll_log_rule_change</code> — who changed which rule, threshold or zone, and when.	Settings → Change log
Delete a client safely	<code>admin_delete_payroll_client</code> raises <code>client_has_riders</code> / <code>system_client</code> rather than deleting a live roster.	Settings → Delete client
Range inside the Kuwait bound	From / To is stitched from monthly snapshots (<code>stitchPayrollRange</code>); each month's days are clipped and concatenated, never re-evaluated.	Range pills → Custom

3. Gold table A — Americana (zone band + orders)

Seeded order, first match wins. Default when nothing matches = **Full day · 12 h**.

#	Conditions	Result	Credits
1	Orders < 1	Absent	0 h
2	Zone is Khiran	Full day · 12 h	12 h
3	Zone band Low and orders < 5	Reduced day · 3 h	3 h
4	Zone band is one of Good, Average and orders < 7	Reduced day · 3 h	3 h
<i>no rule matched → client default</i>		Full day · 12 h	12 h

Input day	Expected rule	Expected status	Expected hours
Khiran · 0 orders · any hours	Rule 1 (before the Khiran exception)	Absent	0
Khiran · 3 orders	Rule 2	Full day	12
Low · 4 orders	Rule 3	Reduced 3 h	3
Low · 5 orders	none → default	Full day	12
Average · 2 orders	Rule 4	Reduced 3 h	3
Good · 6 orders	Rule 4	Reduced 3 h	3
Good · 7 orders	none → default	Full day	12
Not set band · 5 orders	none (no rule matches <code>not_set</code>) → default	Full day	12

The last row is the one to watch: an unmeasured zone must fall to the default, never to Rule 3's "Low".

4. Gold table B — Keeta (hours + orders)

Default when nothing matches = **Actual hours**.

#	Conditions	Result	Credits
1	Hours < 0.5 and orders < 1	Absent	0 h
2	Orders < 3	Absent LO	0 h
3	Hours < 4	Absent LH	0 h
4	Hours < 6 and orders ≥ 6	Half day · 6 h	6 h
5	Hours ≥ 10 and hours ≤ 12 and orders ≥ 6	Actual hours	logged
6	Hours ≥ 10 and hours ≤ 12 and orders < 6	Half day · 6 h	6 h
7	Hours > 12	Full day · 12 h	12 h
no rule matched → client default		Actual hours	logged

Hours	Orders	Expected rule	Expected status	Expected hours
0.0	0	1	Absent	0
0.3	0	1	Absent	0
9.0	2	2	Absent LO	0
3.0	8	3	Absent LH	0
5.0	6	4	Half day	6
5.9	5	none → default	Actual	5.9
11.0	8	5	Actual	11
11.0	4	6	Half day	6
12.0	6	5 (≤ 12 inclusive)	Actual	12
13.0	8	7	Full day	12
8.0	6	none → default	Actual	8

Row 6 is the boundary to be strict about: 5.9 h with 5 orders must fall through to Actual (5.9 h), not to Half day — orders < 6 fails the rule.

5. Gold table C — zone efficiency arithmetic

Zone	Orders	Rider-days	DPD	Efficiency	Band (110 / 70)
A	1,200	40	30.00	130.4%	Good
B	400	25	16.00	69.6%	Low
C	0	0	—	—	Not set

- ❑ Target DPD = $(30.00 + 16.00) \div 2 = \mathbf{23.00}$. Zone C is excluded because it has no rider-days.
- ❑ Zone A = $30 \div 23 \times 100 = \mathbf{130.4\%}$ → Good.
- ❑ Zone B = $16 \div 23 \times 100 = \mathbf{69.6\%}$ → Low. Note it is below 70 — a 70.0 boundary reading must be Average, a 69.6 must be Low.
- ❑ Fail case: if Zone C had been counted as 0, the target would be 15.33 and Zone B would read **104.3%** → Average. If you see Average for Zone B, the empty zone is being counted.
- ❑ Set the override for Zone B to **Good**, recompute, and Zone B must stay Good with an **Edited** tag and the automatic band still visible beside it.
- ❑ Change the client's thresholds to Good ≥ 105 / Average ≥ 60: Zone A stays Good, Zone B becomes **Average** ($69.6 \geq 60$).
- ❑ A zone with a 0 rider-day month must never show 0.00 DPD — it shows an em dash and *Not set*.

6. Gold table D — Required Hours and Attendance Efficiency

Required = (month days – Off Structure) × full-day hours. Efficiency = Actual ÷ Required, uncapped.

Month days	Off Structure	Required h	Actual h	Efficiency
31	2	348	300	86.2%
31	4	324	300	92.6%
30	2	336	336	100.0%
30	2	336	400	119.0%

- Change a rider's Off Structure from 2 to 4 and Required Hours must drop by exactly **48 h** (4×12) while every **day cell stays the same**.
- A rider with no Off Structure row must use the default of 2. The dialog's source line reads *default*.
- A rider with zero check-ins and no leave has Actual = 0 and Efficiency = 0.0%, not blank.
- Efficiency is uncapped — 119.0% must print, not 100%.
- The **Riders at / above 100%** KPI counts riders whose efficiency ≥ 100 , and it must agree with the table row by row.

7. Verify — Admin · A. Shell, range and filters

- Five tabs render: **Payroll** · **Combined Payroll** · **Attendance & Orders** · **Requests** · **Settings**. Direct routes `/en/payroll/combined`, `/en/payroll/attendance-orders`, `/en/payroll/settings` answer with the auth gate (307), not 404.
- **This month / Last month / Custom** pills change the month. Custom refuses a date outside this month and the previous two (*Pick this month or one of the previous two*).
- Apply a custom From / To spanning a month boundary and confirm the day columns are the stitched range and the rider row count does not double.
- **Partner view** pills filter the page; a client with 0 riders in the window is not silently dropped from the count line.
- **Zone category** chips (Good / Average / Low / Not set) filter riders; the summary strip's Good / Avg / Low / Not set counts must match the chips.
- The summary strip reads the client's criteria — *rules use zone + orders* for Americana, *orders + hours* for Keeta.
- Legend chips filter the page and show the share card; clicking the same chip again clears it. The KPI **Riders in payroll** stays the full set while the status chips narrow the page — confirm the intended behaviour against the client's expectation.
- **Clear filters** resets slicers, status and zone category in one go.
- Search matches name, zone, restaurant, AM ID and Driver ID.

8. Verify — Admin · B. Combined grid (the highest-risk area)

- The hint reads *Drag across cells to select · drag the corner handle to fill · Ctrl+C, Ctrl+V, Ctrl+D*.
- Drag a rectangle. The selection bar reads *N cells selected · R riders × D days* with both dimensions correct.
- Drag the **green corner handle** down: the cells it will overwrite show a **dashed preview** before release, and the fill copies the *pattern* of the source block (e.g. 3 h, 12 h, 3 h repeats), not one flat value.
- **Ctrl+C** on a 2×1 block then **Ctrl+V** on a 2×3 block: the copied block **tiles** across the selection. A clipboard smaller than the selection must not paste only once.
- **Ctrl+V** onto a selection whose shape cannot hold the clipboard answers *The pasted block did not match the selected cells*. — not a silent partial write.
- **Ctrl+D** fills down within the selection; **Esc** clears it; **Clear** on the selection bar clears it.
- **Apply with nothing selected** → *Select at least one day first*.
- **Apply with an empty reason** → *A reason is required*. and nothing is written. Re-check the cells afterwards.
- **Keep selected status** leaves every cell exactly as the rules computed it — the dialog says *Every selected cell is already <status>* when they agree.
- Apply a real change (e.g. 6 cells to Half day) and confirm the toast reads *Adjusted 6 day cells* and the KPI **Manual Adjustments** moves.
- The changed cells carry the **orange triangle marker** at the top-end corner. The Attendance & Orders tab shows the same marker on the same day.
- A cell whose day came from a rule shows its **rule pill** (*Rule 3, 3 h*); a day that fell to the client default shows the default; *Automatic result* appears where a cell is unchanged.
- Reload the month (invalidate the snapshot). The adjusted cells must still be the manual value, not the rule value.
- Submit a batch above **2,000 cells** → *Too many cells in one batch*. Nothing is written.
- **Manual changes record** lists date/time, user, rider, day, original (system) value, adjusted value and reason, newest first. Count the rows against the batches you applied.
- With `payroll.manage` removed mid-session, an adjustment attempt is refused with the not-authorized message rather than half-written.

9. Verify — Admin · C. Off Structure

- ❑ The header button opens a modal titled *Off Structure / Contracted OFF days for <month>*, with footer-first layout and Close outside the frame.
- ❑ Save one rider. The status line updates to *Current: N days (manual)* and the toast reads *Off Structure saved*.
- ❑ **Download template** produces the sheet with a Driver ID and an Off days column.
- ❑ Upload a sheet with these rows and check every verdict:

Sheet row	Expected verdict
Valid Driver ID, Off days 3, differs from current	Will apply
Valid Driver ID, Off days = current value	No change
Blank ID cell	Missing ID
The same ID twice	First row evaluated, second Duplicate
Off days abc / 2.5 / -1	Invalid off days
Off days 45 in a 31-day month	Exceeds month
ID that resolves to nobody	Unknown ID
An Employee ID that matches two riders	Ambiguous ID (never a coin-flip)
Month column filled with next month	Wrong month

- ❑ Both a valid **Employee ID** and a valid **Driver ID** must resolve; mixing them in one file is allowed.
- ❑ A sheet where every row is *No change* answers *No Off Structure changes to write. Default OFF = 2 stays implicit.* — and writes nothing.
- ❑ **Apply valid rows** reports *Applied N · skipped M*; the N rows only are visible afterwards with source *bulk upload*.
- ❑ A sheet above **2,000 rows** → *Too many rows in one upload*.
- ❑ A non-spreadsheet file, or a sheet missing the Off days column → *Need a Driver ID / Employee Number column and an Off Days column*.
- ❑ After any apply, the Payroll tab's Required Hours and Efficiency update for exactly those riders, and no day cell moves.

10. Verify — Admin · D. Settings: clients and rules

- ❑ **Americana** and **Keeta** are listed with their criteria ticks (Americana: zone + orders; Keeta: orders + hours) and the SOP rule list for the month.
- ❑ The rule table renders as sentences with the credited hours per row — no raw jsonb anywhere on screen.
- ❑ **Add rule** appends a row; conditions can be added and removed; up / down reorders; an empty condition list never matches.
- ❑ **Save rules** persists and the toast names the count. Reload and the saved order is identical.
- ❑ **Restore SOP starting rules** reinstates the seeded list and toasts *SOP starting rules restored*.
- ❑ Delete a rule and confirm the day it used to decide now falls to the next rule or the client default.
- ❑ **Client criteria**: change Full day hours to 10, save, and every Full day cell must credit 10 h. Revert.
- ❑ **Good must be ≥ Average**. Good 60 / Average 70 is refused with *Good must be at least Average*.
- ❑ Hours outside 0–24 → *Hours must be between 0 and 24*.
- ❑ **Add client** with a duplicate name → *A client with that name already exists*. A blank name → *Give the client a name*.
- ❑ Add a client copying Americana's rules; the new client must hold an independent copy (edit one and the other must not move).
- ❑ **Delete client** with riders assigned → *This client still has riders assigned*. A system client → *System clients cannot be deleted*.
- ❑ Autosave: a criteria change toasts *Saving...* then *Saved*; add / delete / reset toast as well.

11. Verify — Admin · E. Simulator, zones and change log

- ❑ **Try the rules** answers *Decided by the rule "<label>" plus Credited <status> · N h, or No rule matched — the client default applies., or This client has no rules for the month, so the legacy classification applies.*
- ❑ Walk the simulator through every row of Gold table A and B. Every row must match.
- ❑ Set a zone band of **Not set** in the simulator and confirm no Americana rule matches and the default (12 h) is answered.
- ❑ The simulator must never leave a mark — reload Settings and nothing it entered has been saved.
- ❑ **Zones & efficiency** shows Orders · Rider-days · DPD · Target DPD · Efficiency (auto) · Automatic category per zone, with the note that the figures belong to the previous completed month.
- ❑ **Recompute from last month** refreshes the table and keeps every existing override.
- ❑ Override DPD used / target used / final category for one zone, press **Use**, and the row gains **Edited** with the automatic value still shown beside the used one.
- ❑ **Use** on the synthetic (**no zone**) row must be refused.
- ❑ An override of the category changes which rule the riders in that zone match — prove it by switching a zone from Low to Good and re-reading a rider's day.

- ❑ **Change log** carries a row for every action taken above, with When · By · Client · Action · Detail, newest first, and your own name on the rows you caused.

12. Verify — Admin · F. Permissions

Permission set	Expected
payroll.view only	All five tabs readable. No Off Structure button, no Adjust, no Settings save, no Export.
payroll.view + payroll.export	Export to Excel works and downloads the filtered rows.
payroll.view + payroll.manage	Off Structure, adjustments and Settings saves all work. Export still hidden.
All three	Everything available.
No payroll permission	The route is refused; the sidebar item is not offered.

13. Verify — User (rider app inputs)

There is no rider-facing payroll screen in this release and no rider app build to install. Payroll v4 is admin-only. What is tested here is that the three rider-side inputs land correctly on the admin grid. Any tester filing "the app has no payroll page" is filing the wrong ticket.

Rider action	Where it lands on Admin	Expected
Clock in at 08:00, clock out at 17:30	Payroll tab → Actual Hours for that date	9.5 h
Open pickup, complete it, admin verifies	Final orders for that Kuwait day	+1 order, and the client rule reads that number
Raise a Leave request (app shows Pending)	Payroll tab → the day stays <i>unjustified</i>	Unchanged day; rider counted in Riders with unjustified days
Admin approves the Leave request	Payroll tab → the day	OFF, 0 h, rider leaves the unjustified count
Admin hand-adjusts the same approved day	Combined tab → the cell	The adjustment wins, orange marker present
A rider on a client with no saved rules clock-in	Payroll tab → the day	Full day 12 h (legacy), unchanged from before v4

- ❑ Clock-in hours: an open attendance log (clocked in, not yet out) must credit **0 h** for that day, not a running total.
- ❑ A clock-out before the shift start must not add the pre-shift gap.
- ❑ Orders: an *in transit* or *pending* order must not raise the day's orders — only a **verified** order counts toward the rule.
- ❑ Requests: only an **Approved** Leave / Sick / Accident / Vehicle request covers a day. Pending and Under review leave the day unjustified.
- ❑ Arabic locale: the rider sees the ordinary request flow in Arabic; nothing new appears in the app.

14. Regression guardrails — the four that must never break

1. **A client with no saved rules changes nothing.** Its month must be byte-identical to the pre-v4 classification: clocked-in = 12 h, covered = the cover, otherwise Absent.
2. **Operations beats a rule.** An approved Sick day must be Sick · 0 h even when the rule list would have said Full day.
3. **A manual adjustment beats Operations.** Adjusting an approved Sick day to Full day produces Full day, with the marker and the reason recorded.
4. **A recalculation never overwrites an adjustment.** Recompute the month and every hand-adjusted cell keeps its manual value.

15. Automated cover

Suite	Command	Result
Payroll v4	npm run test:payroll	120 / 120 pass across formulas, rules engine, zone metrics, off-structure bulk, rule table model, snapshot
Types and build	npm run typecheck · npm run build	Clean

The engine tests are the SOP's own worked examples and both rule tables from \$5.2 and \$5.3, plus the range prorate, the range stitch, the paste-tile behaviour and the rule-table round-trip. A green suite does not replace \$7–\$12 — it only proves the arithmetic.

16. Sign-off

Block	Pass	Fail	Tester / date	Notes
Shell, range, filters, KPIs	☐	☐		
Gold table A — Americana	☐	☐		
Gold table B — Keeta	☐	☐		
Gold table C — zone banding	☐	☐		
Gold table D — required hours	☐	☐		
Combined grid — select / fill / paste	☐	☐		
Adjustment reason mandatory + audit row	☐	☐		
Off Structure per rider + bulk verdicts	☐	☐		
Settings — clients, rules, validation	☐	☐		
Simulator matches both rule tables	☐	☐		
Zones recompute + override survives	☐	☐		
Change log completeness	☐	☐		
Permissions matrix	☐	☐		
Rider inputs land on the grid	☐	☐		
Four regression guardrails	☐	☐		

Internal QA only. The client-facing document is `client-payroll-v4.pdf`. Do not paste migration filenames, table or RPC names into the client email. Payroll v4 is admin-only — no rider app release is involved.